

# Tournesol: Permissionless Collaborative Algorithmic Governance with Security Guarantees

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# 1. Introduction

- Large-scale algorithms play a major political role in the digital information war — from search engines to chatbots and recommendation systems — subtly shaping public opinion and even geopolitical narratives.  
(*Chesney & Citron 2019; Woolley 2020; Garon 2022*)
- Despite their massive influence, algorithmic information systems are still largely unregulated, opaque, and lacking in accountability — as evidenced by the Facebook Files.  
(*Hagey & Horwitz 2023; Hoang 2020*)

# 1.1 Problem Formulation

- The goal is to define a function  $f$  that transforms data inputs  $(D_1, \dots, D_N)$  from  $N$  user accounts into a scoring vector  $\rho \in \mathbb{R}^A$  over a set of alternatives  $[A] = \{1, \dots, A\}$  — ensuring that  $f$  behaves well and fairly.
- In a **permissionless** system, where anyone can create accounts, security risks arise. We address this with **Lipschitz resilience**, ensuring that no single user (or group using multiple accounts) can exert disproportionate influence on the outcome  $\rho$ .  
(Allouah et al. 2022)
- Many collaborative platforms (e.g., recommendation or voting systems) fit into this framework.

## 1.2 Contributions

- Introduced an end-to-end pipeline for **permissionless, secure collaborative algorithmic governance**, applied to content recommendation.
- Developed new tools like **LipschiTrust** to securely propagate trust and estimate statistics under uncertainty, extending **Lipschitz resilience**.
- Deployed the system on **Tournesol**, gathering 137,000+ comparisons from 7,000+ contributors across 27,000+ videos by August 2023.

## 1.3 Related Work

- Tournesol addresses concerns about **disinformation**, **radicalization**, and **mental health** risks from social media — proposing content prioritization to amplify public interest content.  
(Woolley 2020; Gao et al. 2020; Hoang 2021)
- It builds on prior **algorithmic governance frameworks** like WeBuildAI and preference-based voting agents, generalizing them to content recommendation with a **permissionless and secure** design.  
(Lee et al. 2019; Noothigattu et al. 2018)
- Tournesol uses **trust-based identity** via peer vouching instead of Proof of Work/Stake, drawing from models like **EigenTrust** and ensuring **Lipschitz resilience** for security and fairness.  
(Kamvar et al. 2003; Allouah et al. 2022)

## 2. Interfaces and Data

- Tournesol's governance system relies on two types of input data:
  - (1) contributors' **authentication data**, and
  - (2) their **comparative judgments** between content items.
- These inputs are essential for securing participation and extracting meaningful preferences from the crowd.

## 2.1 Vouch Network

- Contributors are **pretrusted** if their email domains (e.g., @epfl.ch, @who.int) are considered **Sybil-resilient** — i.e., unlikely to allow fake or multiple accounts.
- To improve inclusiveness, Tournesol introduces a **vouching system** where contributors can **vouch for others** as real persons (Proof of Personhood).
- The vouch data captures who vouched for whom, when, and under what criteria — currently limited to **PoP**, but extensible to other trust types in the future.



## 2.2 Comparisons

- Tournesol collects tuples  $(n, a, b, q, r)$ , where contributor  $n$  compares alternatives  $a$  and  $b$  on criterion  $q$ , giving a rating  $r \in [-R_{max}, R_{max}]$  with  $R_{max} = 10$ . A negative  $r$  indicates preference for  $a$  over  $b$ .
- Each criterion  $q$  is treated independently. Comparisons are **anti-symmetric**, and only the **most recent** comparison per pair is retained.
- A contributor's comparisons can be represented as an **anti-symmetric matrix**  $r_{ab}$ , with undefined entries marked as  $\perp$ .

## 2.3 Privacy

- Tournesol allows contributors to **score content privately**, which is important when public support might pose risks (e.g., political or workplace retaliation).
- Only comparisons between **publicly scored** alternatives are included in public datasets; private ones remain hidden and non-auditable.
- As of August 2023, **no differential privacy** is applied — future work aims to address this limitation.  
(Redberg et al. 2023)

### 3. Voting Rights

- Tournesol assigns **permissionless, Sybil-resilient voting rights** by first **computing trust scores** based on contributors' **email verification** and **peer vouching**.
- These trust scores are then used to **determine voting rights** within the governance system.

## 3.1 Trust Scores

- Tournesol introduces a new algorithm, **LipschiTrust**, to assign trust scores.
- LipschiTrust adapts the **EigenTrust** algorithm (*Kamvar et al., 2003*) to enhance **Sybil resilience** in a permissionless environment.

### 3.1.1 Pretrust

- Contributors with verified, **Sybil-resilient email domains** (e.g., @epfl.ch) are assigned a **pretrust score**  $\text{trust}_n^{\text{pre}} = 0.8$ .
- Non-pretrusted contributors receive a **pretrust score** of 0.
- The set of pretrusted contributors is denoted by  $N_{\checkmark}^{\text{pre}}$ .

### 3.1.2 Weighted Vouch Network

- Contributors **explicitly vouch** for others in the PoP network, forming directed edges  $(n, m) \in V_{\text{explicit}}$ .
- A **sink node** with hyperparameter  $V_{\checkmark}^{\text{sink}} = 5$  ensures that trust distribution remains fair, even with few vouches.
- The resulting **weighted vouch network**  $(N, V)$  ensures that the sum of outgoing vouch weights per contributor  $\leq 1$ , making  $V$  **row-substochastic**.

### 3.1.3 The Probabilistic Intuition of LipschiTrust

- LipschiTrust is inspired by **EigenTrust** and **PageRank**, modeling trust as a **random walk** on the vouch network, starting from a uniformly selected pretrusted contributor.
- At each step, with probability  $1 - \beta$ , the walker **resets**; otherwise, it **follows outgoing edges** proportionally to their weights.
- To ensure **Sybil-resilience**, LipschiTrust robustifies the walk by **limiting visits** to over-influential contributors, bounding their maximum impact on trust scores.

### 3.1.4 Defining LipschiTrust

- Initialize trust vector  $\text{tr}_0 \equiv \text{trust}^{\text{pre}} \in \mathbb{R}^N$  based on pretrusted scores.
- At each iteration  $t$ , update trust with:  
$$\text{tr}_{t+\frac{1}{2}} = \text{trust}^{\text{pre}} + \beta V^T \text{tr}_t,$$
where  $\beta \in (0, 1)$  is the vouch decay parameter (Tournesol sets  $\beta = 0.8$ ).
- To prevent **power concentration**, apply **clipping**:  
$$\text{tr}_{t+1,n} = \min \left( \text{tr}_{t+\frac{1}{2},n}, 1 \right).$$



### Proposition 1

There exists a unique trust vector such that  $\|\text{tr}_t - \text{trust}\|_1 \leq |N|\beta^t$  for all  $t$ .

Moreover,  $\text{trust} = \min(\text{trust}^{\text{pre}} + \beta V^T \text{trust}, 1)$ .

### Theorem 1

If  $\text{trust}^{-n}$  is the trust after removing contributor  $n$ 's vouches, then:

$$\|\text{trust} - \text{trust}^{-n}\|_1 \leq \frac{\beta}{1 - \beta} \text{trust}_n^{-n}$$

### Proposition 2

**Vouching cannot decrease trust:** for all contributors  $n$ ,

$$\text{trust} \geq \text{trust}^{-n}$$

### Proposition 3

A contributor's trust score is **positive** if and only if there is a **vouch path** from a pretrusted contributor to them.

## 3.2 Scaling-Calibration Contributors

- Full pairwise scaling (as in Mehestan) is computationally expensive and noisy for contributors with few comparisons.
- To improve accuracy and scalability, Tournesol selects a subset  $N_{\checkmark}^{\text{scaling}}$  of **scaling-calibration contributors**: the  $K_{\text{scaling}}$  contributors with the most comparisons and a trust score above  $\text{trust}_{\text{min}}^{\text{scaling}}$ .
- Tournesol currently uses  $K_{\text{scaling}} = 100$  and  $\text{trust}_{\text{min}}^{\text{scaling}} = 0.1$  to prioritize comparability and efficiency.

### 3.3 Voting Rights for Score Aggregation

- Even contributors with **zero trust** receive a **strictly positive voting right**  $w_a^n > 0$  for alternatives they judged — but influence is limited if many low-trust contributors are involved.
- **Public scorings** are favored through a **privacy penalty**: private votes are weighted by  $w_{\text{private}}^{\text{penalty}} = 0.5$ .
- Voting rights per alternative are computed in four steps, ensuring **fairness** while controlling **security risks**.

### 3.3.1 Cumulative Trusted Score

The total trusted score for an alternative aggregates public scorings and penalized private scorings.

### 3.3.2 Maximal Tolerated Overtrust

The system limits the influence of low-trust voters, setting an overtrust threshold:

$$\text{OverTrust}_a = w_{\times, \text{default}} + f_{\times} \cdot \text{trust}_a$$

with  $w_{\times, \text{default}} = 2$  and  $f_{\times} = 0.1$ .

### 3.3.3 Minimal Voting Right

A minimal voting right  $w_{\min, a}^{\text{public}}$  is computed via dichotomic search to keep overtrust under control.

### 3.3.4 Scoring Voting Rights

Each contributor's voting right is:

-  $\max(\text{trust}_n, w_{\min, a}^{\text{public}})$  if public; penalized if private.

## 4 From Comparisons to Individual Raw Scores

- For each contributor  $n$  and quality criterion  $q$ , their **comparisons**  $r_{ab} \in [-R_{\max}, R_{\max}]$  are normalized into **raw scores**  $\tilde{r}_{ab} \in [-1, 1]$ .
- Since contributors fully control their own comparisons, **security concerns** like Lipschitz resilience are less critical at this stage.
- The process focuses on **standardizing** individual judgments for further aggregation.

## 4.1 A Generalized Bradley-Terry Model

- Tournesol models contributor preferences using a **generalized Bradley-Terry model**: each alternative  $a$  has an implicit true score  $\theta_a^{\text{true}}$ .
- The probability of preferring  $a$  over  $b$  depends on the score difference  $\theta_{ab}^{\text{true}} = \theta_a^{\text{true}} - \theta_b^{\text{true}}$  through a specific probabilistic formula.
- The raw scores  $\theta^{\text{raw}}(r)$  are computed via **Bayesian MAP estimation** with strong convexity guarantees, ensuring **monotonicity** with respect to comparisons.

## Proposition 4: Monotonicity of Raw Scores

The raw scores  $\theta^{\text{raw}}(\tilde{r})$  are **monotonic** with respect to the contributor's comparisons:  
better comparisons for  $a$  lead to higher  $\theta_a^{\text{raw}}$ .

## Computing Uncertainty

- To estimate score **uncertainty**, Tournesol evaluates the **curvature** of the loss around  $\theta^{\text{raw}}$ .
- Specifically, it computes the second derivative (Hessian diagonal ignoring regularization):

$$\Delta\theta_a^{\text{raw}} = \left( - \sum_{b:(a,b) \in AB} \frac{\partial^2}{\partial \theta_a^2} \ln p[r_{ab} | \theta^{\text{raw}}] \right)^{-1/2}$$

- Higher uncertainty reduces a contributor's influence on aggregated global scores.

## 5 Collaborative Score Scaling

- Contributors may use **different implicit scoring scales**, especially when assessing different subsets of alternatives (e.g., top vs. bad content).
- To ensure fair aggregation, contributors' scores must be **collaboratively scaled** to a common reference.
- Tournesol adapts **Mehestan** (Allouah et al., 2022) for secure, permissionless score scaling, with additional modifications tailored to its platform.



## 5.1 Overview of Mehestan

- Mehestan transforms raw scores  $\theta_{n,a}^{\text{raw}}(r)$  into **collaboratively scaled scores**  $\theta_{n,a}^{\text{cs}}(r)$  on a common scale.
- It involves three steps:
  - ❶ **Individual normalization** of each contributor's scores,
  - ❷ **Scaling** scores against others' scores using **Lipschitz-resilient methods**,
  - ❸ **Translation** (shifting) of scores to align contributors.
- The algorithm's resilience is controlled by parameter  $L$ ;  
Tournesol uses  $L = 1/10$ .

## 6 Standardizing, Shifting, Aggregating and Squashing

- After collaborative scaling, scores undergo further **standardization, shifting, aggregation, and squashing** to ensure **human-readability** and **consistency** with unrated alternatives.
- Some steps use a new **Lipschitz-resilient primitive** called **QrQtl**, designed for robust quantile estimation.
- These adjustments refine the scores before final display and aggregation on the platform.

## 6.1 Standardizing the Scores

Collaboratively scaled scores are **standardized** using a Lipschitz-resilient **90%-quantile (QrQtl)** of deviations to control heavy tails.

## 6.2 Shifting the Zero

The zero point is shifted so that **unrated alternatives** fall into the **bottom 15%** ( $q_{\text{shift}}^{\text{zero}} = 15\%$ ), improving interpretability.

## 6.3 Aggregating into Global Scores

Global scores  $\rho_a^{\text{SSS}}$  are computed by **aggregating** contributors' shifted scores  $\theta_{n,a}^{\text{SSS}}$  using a **robust median (QrMed)** with Lipschitz resilience  $L$ .

## 6.4 Squashing for Display

Scores are **squashed** into the bounded interval  $(-100, 100)$  using the mapping:

$$x \mapsto 100 \frac{x}{\sqrt{1 + x^2}}$$

# 7 Conclusion

- We presented an **open-source, permissionless, secure collaborative governance system**, deployed at **tournesol.app**.
- Our pipeline is **functional** but **open to improvements** in areas such as:
  - Handling direct assessments (e.g., likes),
  - Learning volitions,
  - Leveraging expertise and liquid democracy,
  - Enhancing active learning and collaborative filtering,
  - Aligning language models via Tournesol.
- We aim to inspire more **effective, fair, scalable, and interpretable algorithmic governance**.